```
import numpy as np
import matplotlib.pyplot as plt

population_mean = 50
population_std = 10
population_size = 100000

population = np.random.normal(population_mean, population_std,
population_size)

sample_sizes = [30, 50, 100]
num_samples = 1000

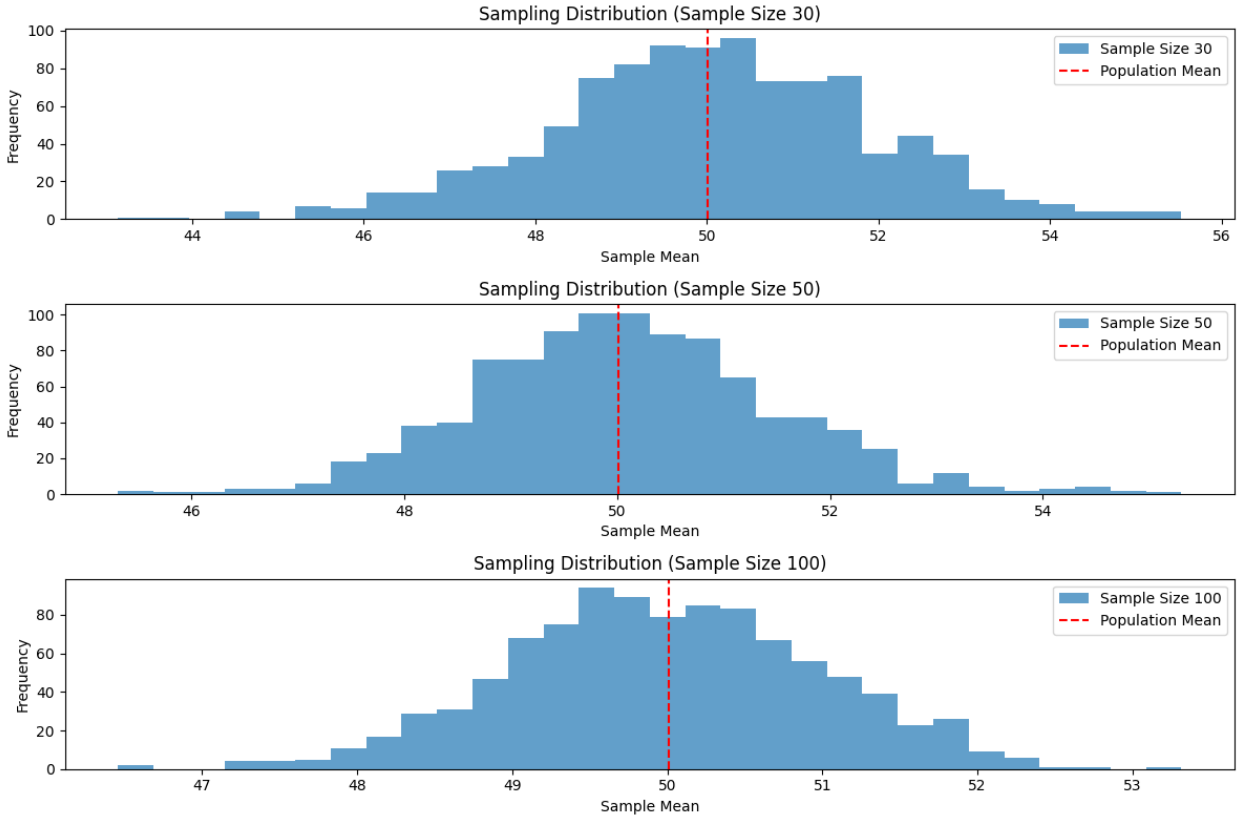
sample_means = {}

for size in sample_sizes:
    sample_means[size] = []
    for _ in range(num_samples):
        sample = np.random.choice(population, size=size,
replace=False)
        sample_means[size].append(np.mean(sample))

plt.figure(figsize=(12, 8))

for i, size in enumerate(sample_sizes):
    plt.subplot(len(sample_sizes), 1, i + 1)
    plt.hist(sample_means[size], bins=30, alpha=0.7, label=f'Sample
Size {size}')
    plt.axvline(np.mean(population), color='red', linestyle='dashed',
linewidth=1.5, label='Population Mean')
    plt.title(f'Sampling Distribution (Sample Size {size})')
    plt.xlabel('Sample Mean')
    plt.ylabel('Frequency')
    plt.legend()

plt.tight_layout()
plt.show()
```


Sampling Distribution (Sample Size 30)
Frequency
Sample Mean
Sample Size 30
Population Mean
0
20
40
60
80
100
44
46
48
50
52
54
56
Sampling Distribution (Sample Size 50)
Frequency
Sample Mean
Sample Size 50
Population Mean
0
20
40
60
80
100
46
48
50
52
54
Sampling Distribution (Sample Size 100)
Frequency
Sample Mean
Sample Size 100
Population Mean
0
20
40
60
80
47
48
49
50
51
52
53